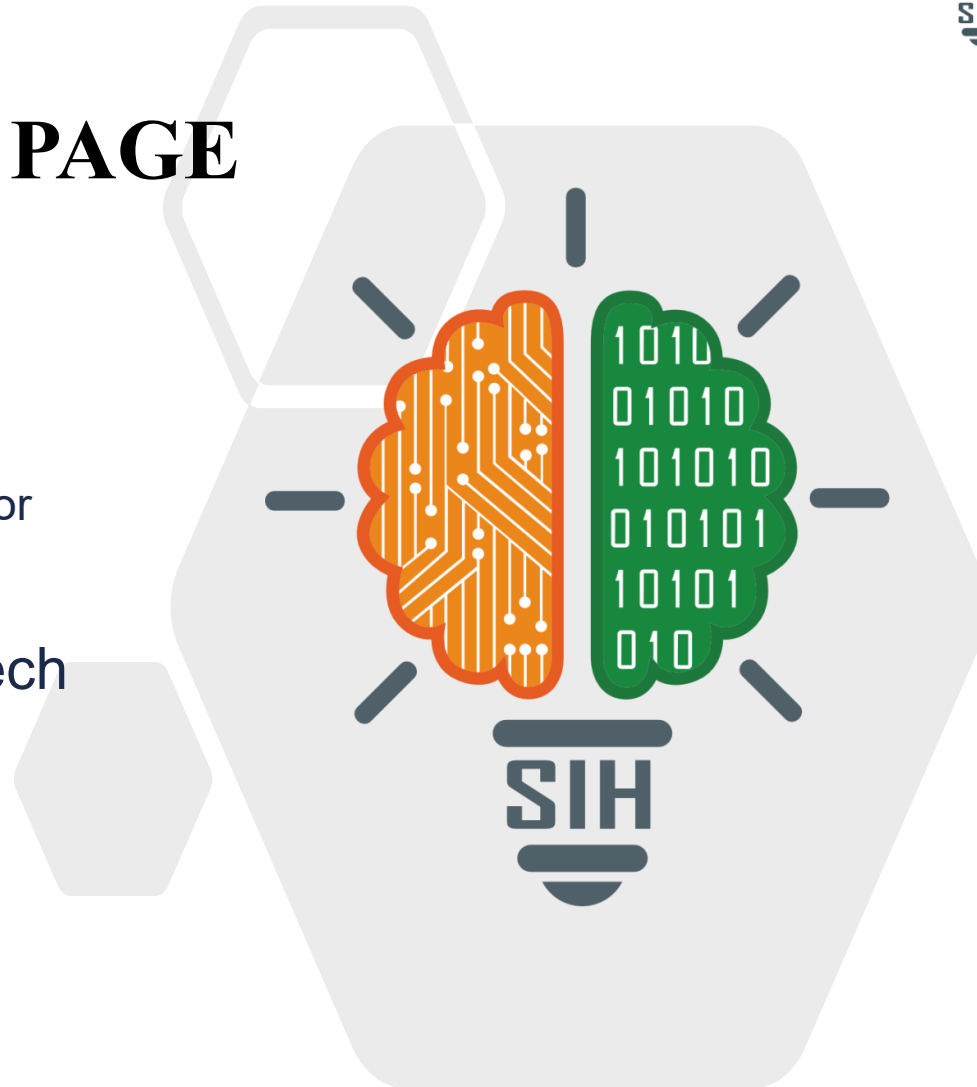


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – SIH26186
- **Problem Statement Title** – AI-Based Predictive Personnel Stress and Welfare Monitoring System for Uniformed Forces
- **Theme** – MedTech / BioTech / HealthTech
- **PS Category** – Software
- **Team ID** – _____
- **Team Name** – Team Name



SAHARA – AI Welfare Intelligence

Predictive Stress & Welfare Monitoring for Uniformed Forces • welfare-first • privacy-first • explainable



Proposed Solution

SAHARA (सहारा = support) is a privacy-first AI platform that fuses HRMS context (duty hours, leave, deployment, night shifts), a 60-second daily check-in and opt-in wearable signals (sleep, HRV) into a 0–100 Welfare Risk Index with a 7-day forecast — explains every score with exact SHAP attributions and puts a specific, evidenced action before the officer who can authorise it. Welfare actions, never disciplinary flags.



How It Addresses the Problem

- Replaces manual observation with continuous early warning — predicts deterioration 7 days ahead with a calibrated escalation probability, before an incident, not after
- Welfare Officer gets ranked cases with the “why”; Command sees only pseudonymised unit aggregates — an individual record returns 403 and is audited
- Recommends interventions ranked by measured counterfactual effect (rest day, break night-shift run, sanction leave, peer support) with SOP citation + what-if simulator



Innovation & Uniqueness

- Consent-gated, tiered data (check-in → HRMS → wearables) with an append-only consent ledger (DPDP Act 2023) — every jawan controls what is shared
- Exact TreeSHAP on every prediction ($\sim 320 \mu s$) + 80 % prediction interval; escalation by 4 inspectable rules — AI ranks, the welfare officer decides
- Stigma-safe by design: HMAC pseudonyms, $k = 10$ suppression, welfare-only RBAC, hash-chained audit — never linked to ACR / promotion / posting
- Model runs in TypeScript (Python parity verified to $1e-13$); zero native dependencies — runs on any laptop, on-prem ready

SAHARA AT A GLANCE

SIGNALS (consent-tiered, opt-in)



HRMS Context

duty hrs · leave · deployment



Daily Check-in

5 questions · 60 seconds



Wearables (opt-in)

sleep · HRV · activity



Predictive Risk Engine

Welfare Risk Index 0–100 · 7-day forecast · P(escalation) · exact SHAP drivers · ranked interventions



Welfare Officer

ranked cases · evidenced actions · what-if simulator



Command / HQ

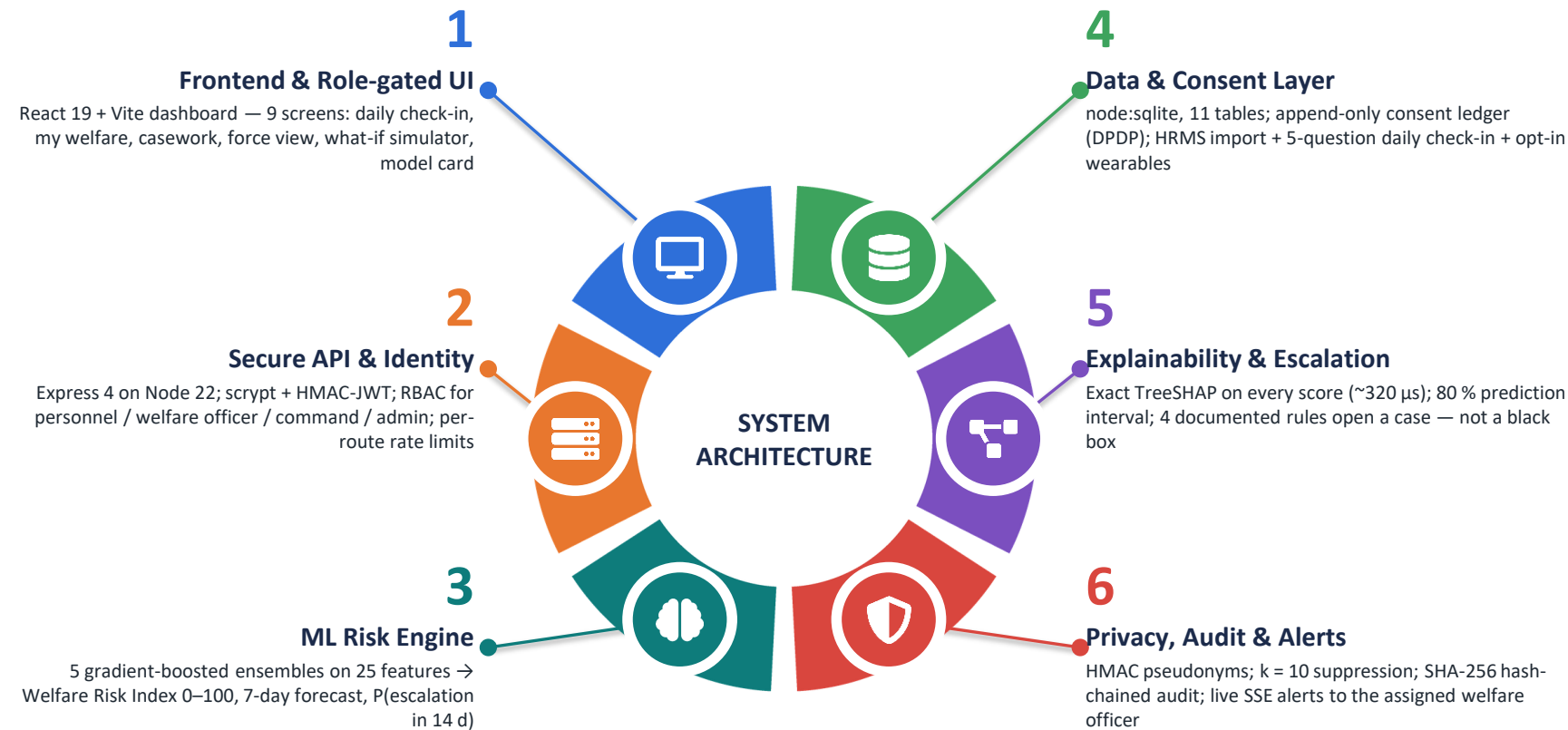
unit aggregates & escalation velocity — pseudonyms only



Privacy Shield

consent ledger · HMAC pseudonyms · $k = 10$ suppression · RBAC · hash-chained audit · DPDP Act 2023

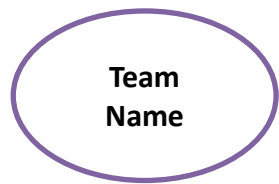
System architecture & technology stack



SAHARA – Core Technology Flow

 Frontend	 React 19 + TypeScript 5.8
	 Vite 6 · Tailwind CSS 4
 UI & Charts	 Motion 12 (reduced-motion aware)
	 Recharts 3 · CVD-safe palette
 Backend	 Node 22 · Express 4
	 SSE live events · rate limits
 ML Training	 Python · scikit-learn
	 pandas · numpy (offline only)
 ML Runtime	 TypeScript GBM inference
	 Exact TreeSHAP ≈ 320 μs
 Database	 node:sqlite (WAL) · 11 tables
	 SHA-256 hash-chained audit
 Security	 scrypt · HMAC-JWT · RBAC
	 HMAC pseudonyms · k = 10
 LLM (optional)	 Gemini 2.0 Flash over SOP RAG
 Build & Deploy	 Vite + esbuild → one Node process
	 Docker · on-prem / NIC MeghRaj

VERIFIED MODEL held-out, split by individual · npm run ml:verify	3.87 WRI MAE	90.3 % band accuracy	0.812 escalation ROC-AUC	0.78 recall (F2-tuned)	79.5 % 80 % PI coverage	≤ 1e-13 TS ↔ Python parity
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FEASIBILITY AND VIABILITY



Feasibility

- ✓ Software-only — no new hardware; wearables strictly optional
- ✓ Leave, duty & transfer data already digitised in CAPF PIS / e-HRMS
- ✓ Working prototype: real trained model (5 GBM ensembles, 25 features), 9 role-gated screens, end-to-end verified
- ✓ Zero native dependencies — one npm run dev; no Python at runtime; SQLite → PostgreSQL swap when scaling
- ✓ Aligned with MHA Task Force (2022) mandate on stress & suicide prevention



Challenges & Risks

- ! Trust & stigma — fear that a risk flag will affect career or posting
- ! Cyber threats on highly sensitive psychological & welfare data
- ! False positives / negatives; cold-start with little labelled data
- ! Low self-reporting in remote, low-connectivity deployments
- ! Cultural & linguistic diversity across forces and units



Mitigation Strategies

- Welfare-only charter: no link to ACR/promotion; jawan can view & contest own score
- scrypt + HMAC-JWT, RBAC, HMAC pseudonyms, hash-chained audit, on-prem hosting, DPDP-compliant
- Human-in-the-loop; threshold tuned on F2 for recall (0.78); model card, calibration & fairness audit in-app
- 60-second, 5-question check-in; HRMS + wearables carry the signal (self-report-only MAE 10.8 vs 3.9)
- Dataset-agnostic pipeline: retrains on a real HRMS CSV without code change; drift (PSI) monitored



₹0 hardware

Software-only platform; opt-in wearables use devices personnel already own



Zero licence cost

100 % open-source stack; indigenous IP retained by MHA / forces



Existing infra

Single Node process; runs on-prem or NIC MeghRaj; air-gap ready



Opt-in & DPDP-ready

Consent-based tiers — no new legal mandate needed to start a pilot

PHASED ROLLOUT ROADMAP

Phase 0 · 0–2 mo

Prototype built on synthetic cohort (2,200 personnel x 12 wk); privacy & security review

Phase 1 · 2–6 mo

90-day pilot in one battalion; validate lead-time & false-positive rate

Phase 2 · 6–12 mo

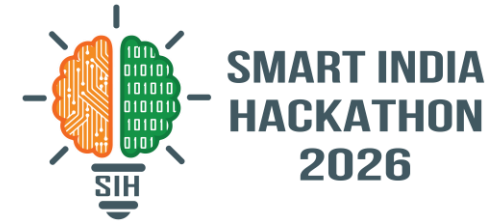
Roll out across one force (e.g., CRPF) with live HRMS integration

Phase 3 · 12–24 mo

All CAPFs, Armed Forces & State Police; cross-force benchmarking

Team
Name

IMPACT AND BENEFITS



730

suicides in CAPFs, NSG & Assam Rifles in 2020–24 (MHA, Rajya Sabha, Dec 2024)

55,500+

voluntary retirements & resignations in the same period — stress and attrition are linked

Reactive → Proactive

SAHARA moves welfare from post-incident response to early, evidence-based prevention

IMPACT

Early identification — of personnel needing welfare support — timely counselling & leave



Fewer incidents — reduction in stress-related incidents, self-harm and fratricide



Balanced workload — data-driven duty rosters, leave planning & transfer decisions



Higher retention — lower attrition / VRS; better job satisfaction & morale



Evidence-based welfare — unit-level trends guide budgets, counsellor deployment & policy



BENEFITS

Social — confidential, destigmatised mental-health support for jawans & families



Organisational — enhanced force readiness, resilience & operational effectiveness



Economic — saves recruitment & training cost of every trained personnel retained



Strategic — indigenous, sovereign welfare-tech for Indian forces (Atmanirbhar Bharat)



Scalable — State Police, NDRF / disaster response, government & corporate HR wellness





Policy & Government Data

1. MHA reply, Rajya Sabha (4 Dec 2024): 730 suicides, 47,891 VRS & 7,664 resignations in CAPFs/NSG/AR, 2020–24
<https://aninews.in/news/national/general-news/730-suicides-amongst-capfs-nsg-and-assam-rifles-during-the-last-five-years-nityanand-rai20241204183618/>
2. MHA Task Force on suicides & fratricide in CAPFs (2022) — service, working & personal stress factors
<https://www.deccanherald.com/india/mha-creates-taskforce-to-prevent-suicides-by-central-armed-police-forces-troops-1056571.html>
3. Digital Personal Data Protection Act, 2023 — MeitY (consent, purpose limitation, data-fiduciary duties)
<https://www.meit.gov.in/data-protection-framework>



Clinical Instruments & Existing Welfare Measures

4. Kroenke et al. (2001) PHQ-9 depression scale; Spitzer et al. (2006) GAD-7 anxiety scale — validated screening tools
<https://pubmed.ncbi.nlm.nih.gov/11556941/>
5. Cohen et al. (1983) Perceived Stress Scale (PSS-10); WHO ICD-11 burn-out definition (2019)
<https://www.who.int/news/item/28-05-2019-burn-out-an-occupational-phenomenon-international-classification-of-diseases>
6. Tele-MANAS (MoHFW) national tele-mental-health helpline 14416 — counselling referral pathway; CRPF ‘Chaupal’ & buddy system
<https://telemanas.mohfw.gov.in/>



Research

7. Schmidt et al. (2018) WESAD — wearable stress & affect detection dataset (ICMI)
<https://dl.acm.org/doi/10.1145/3242969.3242985>
8. Lundberg et al. (2020) TreeSHAP — exact local explanations for tree ensembles (Nature Machine Intelligence)
<https://www.nature.com/articles/s42256-019-0138-9>
9. Friedman (2001) Greedy function approximation: a gradient boosting machine (Annals of Statistics)
<https://projecteuclid.org/euclid.aos/1013203451>



Technology & Standards

10. Pedregosa et al. (2011) scikit-learn: Machine Learning in Python (JMLR) — training stack
<https://jmlr.org/papers/v12/pedregosa11a.html>
11. NIC MeghRaj Government Cloud & CERT-In security guidelines for hosting sensitive data
<https://cloud.gov.in/>
12. Android Health Connect API — consent-based access to wearable HRV, sleep & activity data
<https://developer.android.com/health-and-fitness/guides/health-connect>